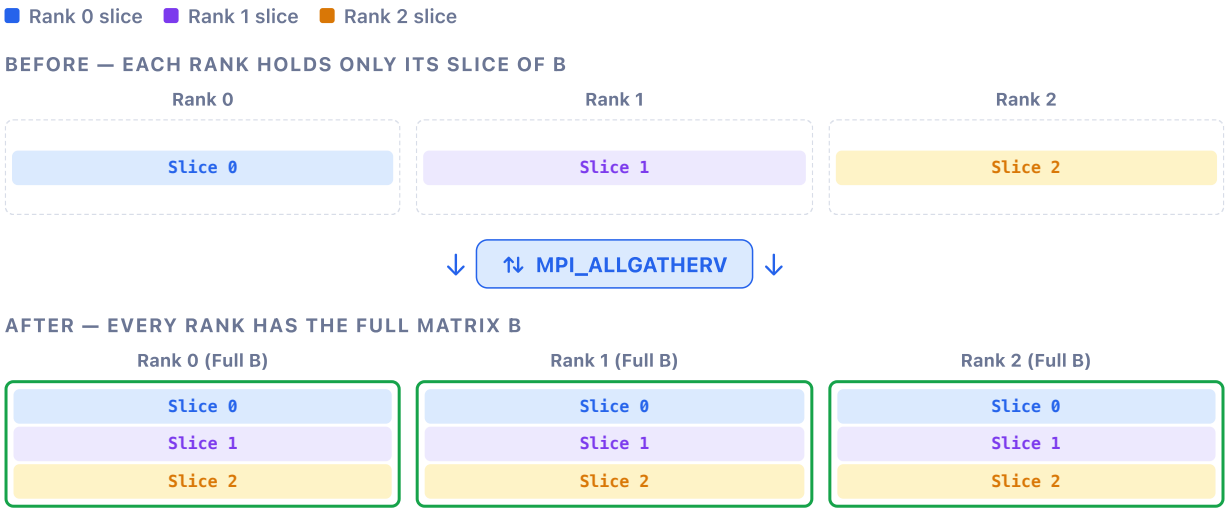
Slide 3: Communication — MPI_Allgatherv

Speaker: Ehtesham Faraz

Collective Operation Flow

Gathering partial slices into full replicas



Implementation Insight

Decoupling communication from computation: completing data replication up-front lets the subsequent $O(N^3)$ matrix multiply proceed with zero network latency.

Strategy Details

Matrix B Replication Logic

Requirement

Every rank must possess a complete copy of Matrix B to compute its row slice of result Matrix C.

Solution

`MPI_Allgatherv` is used. Each rank initialises a unique slice of B, then the collective operation broadcasts all slices to all ranks simultaneously.

Trade-off Analysis

- ✅ **Pro:** Zero communication during the main compute phase.
- ⚠️ **Con:** Memory per node scales as $O(N^2)$. Acceptable for $N \leq 8000$ (~500 MB).

Variable-Size Displacement

Because N may not divide evenly by P, `MPI_Allgatherv` and `MPI_Gatherv` use per-rank `recvcounts` and `displs` arrays.

Rank	recvcounts[i]	displs[i]
0	$rows_0 \times n$	0
1	$rows_1 \times n$	$rows_0 \times n$
...	...	...

</> mpi_matmul.c — communication

```
/* Gather all B slices → full B on every rank */
MPI_Allgatherv(local_B_slice, my_rows * n, MPI_DOUBLE,
               B, recvcounts, displs, MPI_DOUBLE,
               MPI_COMM_WORLD);
free(local_B_slice); /* no longer needed */

/* Gather result slices → full C on Rank 0 */
MPI_Gatherv(local_C, my_rows * n, MPI_DOUBLE,
            C, recvcounts, displs, MPI_DOUBLE,
            0, MPI_COMM_WORLD);
```